American Computer Science League

2023 Finals ● Short Problems Solutions ● Intermediate/Classroom Division

1. Boolean Algebra

$(\overline{B + C})(\overline{A}B + \overline{C})(\overline{A + B + C})$
$= \overline{B}\,\overline{C}(\overline{A}B + \overline{C})(\overline{A}\,\overline{B}\,\overline{C})$
$= \overline{A}\,\overline{B}\,\overline{C}(\overline{A}B + \overline{C})$
$= \overline{A}\,\overline{B}\,\overline{C}B + \overline{A}\,\overline{B}\,\overline{C}$
$= \overline{A}\,\overline{B}\,\overline{C}$

B. $\overline{A}\,\overline{B}\,\overline{C}$

2. Boolean Algebra

$\overline{A(\overline{BC} + C)}\ (\overline{B\overline{C}} + A) = \overline{A(\overline{B} + \overline{C} + C)}(\overline{B} + \overline{\overline{C}} + A)$
$= \overline{A}\,1(A + \overline{B} + C)$
$= \overline{A}A + \overline{A}\,\overline{B} + \overline{A}C$
$= \overline{A}(\overline{B} + C)$

To be TRUE $\overline{A} = 1$ and $\overline{B} + C = 1$. So $A = 0$.
$\overline{B} + C = 1$ for 3 different values. Therefore 3 ordered triples make the expression TRUE: (0,0,0), (0,0,1), and (0,1,1)

B. 3

3. Bit-String Flicking

NOT (010111 XOR 011011 OR 001010 AND 111010)
= NOT (010111 XOR 011011 OR 001010)
= NOT (001100 OR 001010)
= NOT 001110
= 110001

B. 110001

4. Bit-String Flicking

(LSHIFT-1 (RCIRC-2 11011)) OR X = 11110
Let X = abcde.
LHS = (LSHIFT-1 (RCIRC-2 11011)) OR X
= (LSHIFT-1 (RCIRC-2 11011)) OR abcde
= (LSHIFT-1 11110) OR abcde
= 11100 OR abcde
= 111de
LHS = RHS → 111de = 11110
a = *, b = *, c = *, d = 1, e = 0
Therefore there are 8 solutions: ***10

D. 8

5. Recursive Functions

$f(1) = 3$
$f(2) = 2 * f(2-1) - 1 = 2 * f(1) - 1 = 2 * 3 - 1 = 5$
$f(3) = 2 * f(3-1) - 1 = 2 * f(2) - 1 = 2 * 5 - 1 = 9$
$f(4) = 2 * f(4-1) - 1 = 2 * f(3) - 1 = 2 * 9 - 1 = 17$
$f(5) = 2 * f(5-1) - 1 = 2 * f(4) - 1 = 2 * 17 - 1 = 33$

You could continue this way until $f(11)$ or look at the sequence of numbers and look for a pattern.

3 5 9 17 33 .....
2^1+1 2^2+1 2^3+1 2^4+1 2^5+1 ..... 2^n+1
So $f(11) = 2^{11}+1 = 2048 + 1 = 2049$

C. 2049

6. Recursive Functions

$f(24, 6) = f([24/2], 6-1) + 3 = f(12, 5) + 3 = -29 + 3 = -26$
$f(12, 5) = f([12/2], 5-1) + 3 = f6, 4) + 3 = -32 + 3 = -29$
$f(6, 4) = 2 * f(6+1, [4/3]) = 2 * f(7, 1) = 2 * (-16) = -32$
$f(7, 1) = 2 * f(7+1, [⅓]) = 2 * f(8, 0) = 2 * (-8) = -16$
$f(8, 0) = 2 * f(8+1, [0/3]) = 2 * f(9, 0) = 2 * (-4) = -8$
$f(9, 0) = 2 * f(9+1, [0/3]) = 2 * f(10, 0) = 2 * (-2) = -4$
$f(10, 0) = f([10/2], 0-1) + 3 = f(5, -1) + 3 = -5 + 3 = -2$
$f(5, -1) = 5 * (-1) = -5$

C. -26

7. Digital Electronics

The Boolean expression for the circuit is: $(A(A + B)) \oplus \overline{(B + C)}$

$(A(A + B)) \oplus \overline{(B + C)} = (AA + AB)) \oplus (\overline{B}\,\overline{C})$
$= (A + AB) \oplus (\overline{B}\,\overline{C})$
$= A \oplus (\overline{B}\,\overline{C})$
$= \overline{A}\,\overline{B}\,\overline{C} + A(\overline{\overline{B}\,\overline{C}})$
$= \overline{A}\,\overline{B}\,\overline{C} + A(B + C)$
$= \overline{A}\,\overline{B}\,\overline{C} + AB + AC$

If $A = 0$, $\overline{A}\,\overline{B}\,\overline{C} + AB + AC = \overline{B}\,\overline{C}$. This is TRUE for (0, 0, 0).
If $A = 1$, $\overline{A}\,\overline{B}\,\overline{C} + AB + AC = B + C$. This is TRUE for 3 ordered triples: (1,0,1), (1,1,0) and (1,1,1) for a total of 4.

C. 4

8. Digital Electronics

The Boolean expression for this circuit is: $\overline{A(AB + \overline{B + C}) + C}$

$$\begin{aligned}
\overline{A(AB + \overline{B + C}) + C} &= \overline{A(AB + \overline{B}\,\overline{C}) + C} \\
&= \overline{AB + A\overline{B}\,\overline{C} + C} \\
&= \overline{A\,B}\;\overline{A\,\overline{B}\,\overline{C}}\;\overline{C} \\
&= (\overline{A} + \overline{B})(\overline{A} + B + C)\,\overline{C} \\
&= (\overline{A} + \overline{B})(\overline{A}\,\overline{C} + B\overline{C} + C\overline{C}) \\
&= \overline{A}\,\overline{C} + \overline{A}B\overline{C} + \overline{A}\,\overline{B}\,\overline{C} + B\overline{B}\,\overline{C} \\
&= \overline{A}\,\overline{C} + \overline{A}B\overline{C} + \overline{A}\,\overline{B}\,\overline{C} \\
&= \overline{A}\,\overline{C}\,(1 + B + \overline{B}) \\
&= \overline{A}\,\overline{C}
\end{aligned}$$

D. $\overline{A}\,\overline{C}$

9. Prefix-Infix-Postfix

$$\begin{aligned}
+\ \$\ 7\ 3\ -\ *\ \$\ 9\ 7\ 3\ 5 &= +\ (\$\ 7\ 3)\ -\ *\ (\$\ 9\ 7)\ 3\ 5 \\
&= +\ 4^2\ -\ *\ 2^2\ 3\ 5 \\
&= +\ 16\ -\ (*\ 4\ 3)\ 5 \\
&= +\ 16\ (-\ 12\ 5) \\
&= +\ 16\ \ 7 \\
&= 23
\end{aligned}$$

B. 23

10. Prefix-Infix-Postfix

$$\begin{aligned}
&\frac{(A+B)(A-B)}{B} \;-\; \frac{B(4-A)}{A} \\
&= \left(\frac{((A+B)*(A-B))}{B}\right) - \left(\frac{(B*(4-A))}{A}\right) \\
&= (((A + B) * (A - B))/B) - ((B * (4 - A))/A) \\
&= (((AB +) * (AB -))B/) - ((B(4A -) *)A/) \\
&= (AB + AB -* B/) - (B4A -* A/) \\
&= AB + AB -* B/B4A -* A/ -
\end{aligned}$$

D. AB+AB−*B/B4A−*A/−

11. Computer Number Systems

$$\begin{aligned}
52723_8 &= 101\ 010\ 111\ 010\ 011_2 \\
&= 101\ 0101\ 1101\ 0011_2 \\
&= \ 5 \quad 5 \quad D \quad 3_{16}
\end{aligned}$$

B. 55D3

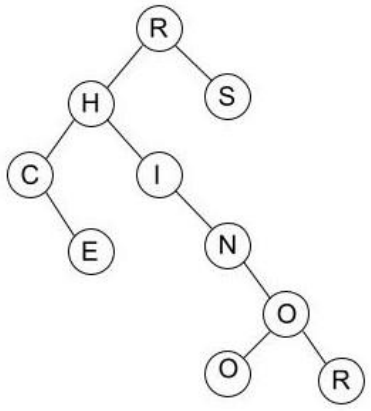
12. Computer Number Systems

$1ABC_{16} = 1 * 16^3 + 10 * 16^2 + 11 * 16 + 12$
$= 1 * 4096 + 10 * 256 + 11 * 16 + 12$
$= 4096 + 2560 + 176 + 12$
$= 6844$

D. 6844

13. Data Structures

The binary search tree for **RHINOCEROS** is:



The internal path length = 2*1 + 2*2 + 2*3 + 4 + 2*5 = 26.

B. 26

14. Data Structures

A stack is LIFO (last in first out). This is described by: d) a spring-loaded device where the ball on top is retrieved and when returned each ball is put on top by pushing the other balls further down

D. d

15. Graph Theory

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

There are 10 pairs of vertices with no path of length 2 which are AB, BA, CB, CD, CE, DA, DB, DC, EA and EE if done by inspection.

C. 10

16. Graph Theory

There are 8 cycles from A. They are: ABA, ABDCA, ABEDCA, ADCA, ADEBA, AEBA, AEBDCA, and AEDCA

C. 8

17. What Does This Program Do?

This program finds the number of swaps needed to put the list in sorted order. The two outside loops find the location of each number, 0 to 9, in the current array. The inside loop swaps the values in the locations between where the number should be and where it currently is and counts how many swaps are made. The following table shows the resulting array and the swaps made:

x	array	count
original	5 8 6 2 9 3 7 0 1 4	0
0	0 7 3 9 2 6 8 5 1 4	4
1	0 1 5 8 6 2 9 3 7 4	8
2	0 1 2 6 8 5 9 3 7 4	10
3	0 1 2 3 9 5 8 6 7 4	12
4	0 1 2 3 4 7 6 8 5 9	15
5	0 1 2 3 4 5 8 6 7 9	17
6	0 1 2 3 4 5 6 8 7 9	18
7	0 1 2 3 4 5 6 7 8 9	19
8	0 1 2 3 4 5 6 7 8 9	19

D. 19

18. LISP

(SETQ M ‘(1 (2 3) (4 (5) (6 7))))
(CDR (REVERSE (CAR (REVERSE (CDR M)))))
= (REVERSE (CDR ‘(1 (2 3) (4 (5) (6 7)))))
= (CDR (REVERSE (CAR (REVERSE ‘((2 3) (4 (5) (6 7)))))))
= (CDR (REVERSE (CAR ‘((4 (5) (6 7)) (2 3)))))
= (CDR (REVERSE ‘(4 (5) (6 7))))
= (CDR ‘((6 7) (5) 4))
= ((5) 4)

B. ((5) 4)

19. FSAs and Regular Expressions

The names must satisfy the following regular expression:

[^bru] [a-iro] [\D]* [a-n]* [^mr] [\d]*

a) asher - s fails at [a-iro]
b) aurora - u fails at [a-iro]
c) c++ - + fails at [a-iro]
d) jo-ann satisfies it
e) trish - satisfies it
f) noah2 satisfies it
g) a1b2c3 - 1 fails at [a-iro]
h) violet - satisfies it
i) william - fails at [^mr]
j) eleanor - l fails at [a-iro]
k) yanni satisfies it
l) lee satisfies it

A. d, e, f, h, k, l

20. Assembly Language

The outside loop divides the number 20 by each of its prime factors until the value of n = 1. The inside loop checks if the numbers 2, 3, 4, 5, 6, ... go into the number evenly. For each one that it finds, the number is divided by that factor. 20/2 = 10 so 10/2 = 5, and 5/5 = 1. If the factor does go into the number evenly, which is true for 2 and 5, it keeps track of the largest value which is 5.

C. 5